OTP

INDUSTRIAL AOI & MACHINE VISION PLATFORM

OTPSoftware.Core • OTPSoftware.App • OTPSoftware.Tests

.NET 10.0 | C# 14 | WPF MVVM | EN | 中文 | VI

TECHNICAL REPORT & FACTORY OPERATIONS USER GUIDE

OTP Log Monitor & Optical Inspection System

Real-time multi-machine production monitoring, MES optical spec verification, and AOI camera defect inspection platform (**MMAPI120 / MMAPI110 / N-Machine Extensible**).

60 FPS
Real-time log tailing (2s poll) & 30,000-unit RAM window

Zero-OOM
Async WIC decode avoiding ~1 GB RGBA raw PNG spikes

3 Languages
Instant runtime switch EN / Tiếng Việt / 中文 without restart

REPORT SCOPE (14 SLIDES)

01 **System Overview & Software Architecture**
Functional Core / Imperative Shell, Zero-Allocation & Zero-OOM

02 **Shop-Floor Operations Manual**
Log Reading, Real-Time Tail, Faulty PMNT & Pareto Charts

03 **Wallboard F11 & MES / Camera Defect Viewer**
Live NG Spotlight, MES Spec Evaluation & Side-by-Side Cams

04 **Configuration, Single-File Rollout & Troubleshooting**
settings.json, Self-Contained x64, Hotkeys & Smoke Check

Release Version: **Version 1.0 (win-x64 Single-File Self-Contained)**

Target Environment: **Windows 10 (1809+) / Windows 11 x64**

Release Date: **2026-11-14**

1. Executive Overview & Shop-Floor Challenges Solved

Production Ready • 24/7 Line Operation

MISSION: **OTP Log Monitor & Optical Inspection System** provides engineers and line leaders with a unified command center to track shift output, First Pass Yield (**FPY**), trigger immediate consecutive-NG Jig (**PMNT**) alarms, and inspect both **MES** optical measurements and **AOI** camera images in one console.

BEFORE Shop-Floor Operational Bottlenecks

- **File Locks & Dropped Lines on Log Rotation:** Active `OTP_1.log` and `MES_1.log` files are continuously written over SMB; naive editors risk sharing violations or lose records during log rotation (`OTP_1.1.log ...`).
- **UI Freezes on Heavy Shift Histories:** Shifts exceeding 200,000+ log lines and hundreds of MBs of MES XML payloads stall traditional text viewers and ad-hoc scripts for minutes.
- **Out-of-Memory Crashes Opening Raw Images:** Sensor frames `photo/Main*.png` at `17701 × 14208 px` (~50 MB disk / **~1 GB uncompressed RGBA** in RAM) rapidly crash 8–16 GB shop-floor PCs.
- **Fragmented Jig & Camera Lookup:** Technicians had to manually cross-check defect codes, SMB logs on 110/120 hosts, and four separate camera IP shares (`A1/A2/B1/B2`).

V1.0 SOLUTION Core Engineering Capabilities

- **Non-Locking SMB Tail Reader:** Opens logs with `FileShare.ReadWrite | FileShare.Delete` plus `64-bit FNV-1a` fingerprinting to detect rotation transparently with zero missed or duplicate lines.
- **Millisecond Reverse MES XML Scanner:** Scans backwards in `4 MB` pooled byte chunks (`ArrayPool<byte>`), decoding only the matched `<BODY>` record and evaluating against optical specs instantly.
- **Zero-OOM Industrial Imaging Engine:** Prioritizes processed `piresult/AlgIntegrate.jpg` & `defectPicture/` crops via `DecodePixelWidth` + `BitmapImage.Freeze()`, cutting per-image RAM from ~750 MB down to **3–8 MB**.
- **All-in-One Wallboard & Defect Viewer:** Unifies 5 KPI cards, 3 interactive charts, **F11** TV Wallboard, **Live NG Spotlight**, consecutive-NG Jig alarms, and scheduled `08:30 / 20:30` shift exports.

Default Line Topology:
Extensible to N machines in `settings.json`

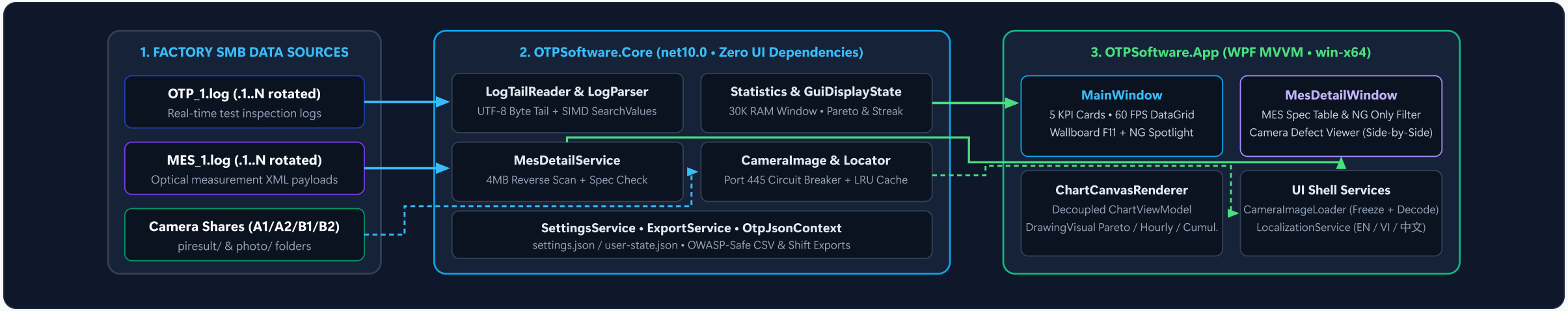
MMAPI120 (`.206`)
Left Cam `B1 (.203)` • Right Cam `B2 (.204)`

MMAPI110 (`.205`)
Left Cam `A1 (.201)` • Right Cam `A2 (.202)`

Auto Shift Export:
Day `08:30–20:30` • Night `20:30–08:30`

2. Solution Architecture (Functional Core – Imperative Shell)

Clean Architecture • .NET 10 / C# 14



Three-Project Structure in `OTPSoftware.sln`

Project	Target Framework	Architectural Role & Key Components
OTPSoftware.Core	net10.0 C# 14	**Pure Functional Core (0 WPF dependencies):** Non-locking SMB log tailer, `ReadOnlySpan<char>` parser, C# 14 `extension` filtering block (`UnitCollectionExtensions`), image locator & MES spec evaluator, `LruMemoryCache` , automatic shift export, and AOT-ready `OtpJsonContext` .
OTPSoftware.App	net10.0-windows WPF MVVM	**Imperative Shell (Presentation Layer):** UI-decoupled ViewModels (`MainViewModel` , `SpotlightViewModel` , `MesDetailViewModel` , `ChartViewModel`), hardware-accelerated `ChartCanvasRenderer` , async Bitmap lifecycle loader, and live 3-language `DynamicResource` switcher.
OTPSoftware.Tests	net10.0-windows xUnit Suite	**Automated Regression Test Suite:** Validates log rotation & oversized line handling, reverse MES XML lookup, LRU eviction invariants, OWASP CSV formula neutralization, and 100% resource key parity across `Strings.{en,vi,zh}.xaml` .

3. High-Performance Engine: Zero-Allocation & Zero-OOM Imaging

Memory Efficiency & SMB Resilience

01 Zero-Allocation Log Tailing & FNV-1a Rotation

- **Non-Blocking File Sharing:** `LogTailReaderService` opens SMB logs with `FileShare.ReadWrite | FileShare.Delete`, never blocking tester writes or log rolling.
- **Byte-Level UTF-8 Splitting:** Streams directly into pooled buffers (`ArrayPool<byte>`) and splits on `\n` bytes before string materialization, eliminating LOH GC spikes.
- **Rotation Continuity:** Stores `64-bit FNV-1a` head/tail fingerprints; when `OTP_1.log` rotates to `OTP_1.1.log`, drains remaining old tail lines before attaching to the new file.
- **SIMD Noise Filtering:** `LogParserService` uses `SearchValues<string>` to skip irrelevant lines at ~192 bytes per 1,000 log lines.

02 4 MB Reverse MES XML Scanner & Spec Check

- **Reverse Chunk Scanning:** `MesDetailService` scans backwards from EOF of `MES_1.log` in `4 MB` pooled blocks to locate the latest panel run in milliseconds.
- **Selective UTF-8 Decode:** Searches UTF-8 serial patterns directly on `ReadOnlySpan<byte>`; allocates strings only for the matched `<BODY> ... </BODY>` payload (~1–9 KB).
- **Zero-Split Reader:** Parses pipe-delimited (`|`) parameter strings via stack-only `PipeSectionReader` and looks up specs using `.GetAlternateLookup<ReadOnlySpan<char>>()` + `[GeneratedRegex]`.

03 Industrial Zero-OOM Camera Engine

- **Smart Format Priority:** Automatically loads lightweight processed JPGs (`piresult/AlgIntegrate.jpg` ~9–12 MB and `defectPicture/*.jpg`) instead of raw `photo/Main*.png` (~50 MB / ~1 GB RGBA).
- **WIC Hardware Downsampling:** Decodes via `FileStream` into `BitmapImage` with `BitmapCacheOption.OnLoad` and `DecodePixelWidth` (**360 px** Spotlight, **1600 px** Detail Viewer), capped at **4096 px** in `1:1 Raw` mode.
- **Thread-Safe Freezing:** Always calls `bitmap.Freeze()` on background workers and cancels superseded SMB tasks via `CancellationTokenSource` on rapid navigation.

04 10s SMB Circuit Breaker & 30K Sliding Window

- **Instant Offline Fail-Fast:** `ImageLocatorService` probes TCP port 445 (600–1200 ms timeout); offline hosts trip a **10-second Circuit Breaker** so the UI never freezes on Windows SMB timeouts.
- **Zero-Allocation Node-Reusing LRU:** Thread-safe `LruMemoryCache<TKey, TValue>` recycles linked nodes (0 steady-state bytes on hit/update), capped at 24–100 recent entries.
- **Bounded 30K RAM Window & UI Virtualization:** Retains the latest `30,000` records (trimmed in batches of 2,000) paired with `VirtualizingStackPanel` for steady 60 FPS scrolling.

4. Camera Directory Structure & 3-Tier Lazy Search Strategy

A1 / A2 (MMAPI110) • B1 / B2 (MMAPI120)

Unit Inspection Folder Naming Convention

`[Status]_[Serial]_[Timestamp]_[Station]`

Status Inspection outcome: `OK` or `NG`

Serial Panel serial / barcode from MES (e.g. `TL687304F0ED`)

Timestamp Capture timestamp token (e.g. `1113091205`)

Station Camera channel: `A1`, `A2` (110) / `B1`, `B2` (120)

Examples: `OK_TL687304F0ED_1113091205_A1` • `NG_TL693874Q0BB_1114083012_B1`

3-Tier Lookup Algorithm (Lazy Enumeration)

1. **Log Timestamp Date Folders:** Probes standard date formats matching the unit timestamp (`yyyyMMdd`, `yyyy-MM-dd`, `yyyy_MM_dd`, `yyyy-MM\yyyy-MM-dd`).
2. **Midnight Handover Tolerance:** Also checks adjacent day folders (D-1 / D+1) to catch panels captured across the 00:00 shift boundary.
3. **Root & Top-15 Latest Folders:** Uses streaming `Directory.EnumerateDirectories` so SMB shares exceeding 50,000+ folders respond in milliseconds.

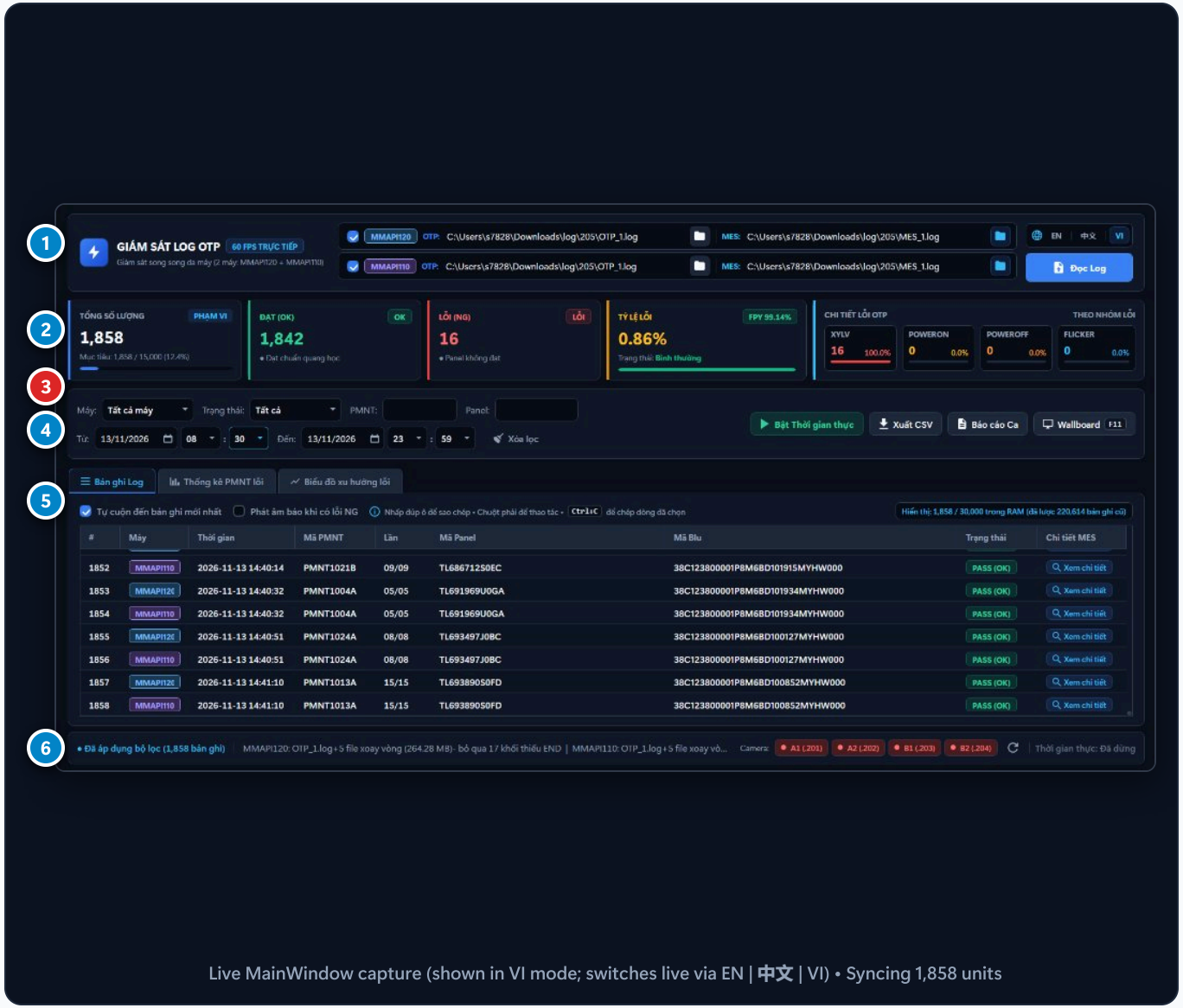
Standard SMB Camera Share Folder Tree

```
\\192.168.3.203\e\LOG\
└── NG_TL687304F0ED_1113091205_A1\
    │
    ├── piresult\                    [DEFAULT LOAD PRIORITY]
    │   ├── AlgIntegrate.jpg     Composite algorithm image (~9–12 MB)
    │   ├── Main_*_L255.jpg      Full-frame L255 fallback JPG
    │   ├── defectPicture\       ROI crops by defect code (XYLV/POWERON ... )
    │   │   └── XYLV_ROI1_001.jpg
    │   └── classifyPicture\     Secondary defect classification thumbnails
    │       └── Classify_001.jpg
    │
    ├── photo\                       [ON-DEMAND 1:1 RAW MODE ONLY]
    │   ├── Main_*_L0.png        Raw luminance frame L0
    │   ├── Main_*_L255.png      Full 17701×14208 px sensor frame (~50 MB)
    │   └── Sub_RGB_*.png        Raw RGB / grayscale channel subframes
    │
    └── debugData1\                  [VISION ALGORITHM METADATA]
        ├── roi.dat              ROI coordinates & alignment matrices
        └── defect.dat           Defect vectors & pixel cluster coordinates
```

PRACTICAL GAIN: When a panel fails with `XYLV`, Spotlight automatically prioritizes crops matching `XYLV` inside `defectPicture/` before falling back to `AlgIntegrate.jpg`, focusing operators straight on the defect ROI.

5. Main Dashboard Anatomy (MainWindow)

6 Primary Functional Zones



Live MainWindow capture (shown in VI mode; switches live via EN | 中文 | VI) • Syncing 1,858 units

1. **Source & Language Header Bar:** Enable/disable machines, browse `OTP …` / `MES …` paths, switch `EN | 中文 | VI` live, and click **Read Log**.

2. **5 Real-Time KPI Cards:** TOTAL UNITS (with 15,000 shift target progress), PASS (OK), FAIL (NG), DEFECT RATE / FPY (color-coded *Green <2% / Amber 2–5% / Red ≥5%*), and OTP defect breakdown.

3. **Consecutive NG / Defect Alert Banner:** Auto-reveals below KPIs on new defects or when a Fixture reaches **3 consecutive NGs** (auto-hides when clear).

4. **Filter & Action Bar:** Instant filters by Machine, Status, PMNT, Panel, and `From/To` dates; toggle **Start Real-time (2s)**, Export CSV, Shift Report, or **Wallboard F11**.

5. **3 Analytics Tabs & Grid:** Switch smoothly between *Log Records*, *Faulty PMNT Statistics*, and *Defect Trend Chart*; click **View Details** for MES & Camera inspection.

6. **Diagnostics Status Footer:** Shows rotated log counts, skipped incomplete blocks lacking `END` tags, and live TCP 445 SMB health LEDs per camera.

6. Quick Start Workflow: "Read Log" vs "Start Real-time"

Start-of-Shift SOP

1 Launch & Select Machines

Open `OTPSoftware.App.exe`. Config loads from `config/settings.json` and restores machine toggles (`MMAPI120`, `MMAPI110`) from `user-state.json`.

2 Verify Log File Paths

Check `OTP_1.log` & `MES_1.log` boxes on the header. After editing paths or browsing via `OTP ...` / `MES ...`, press `Enter` to persist.

3 Load Baseline Shift Logs

Click **Read Log** to load the active log and all rotated logs matching the selected `From / To` window (up to 30,000 latest units in RAM).

4 Enable Real-time (2s Poll)

Click **Start Real-time** to tail active log files in parallel every `2 seconds`, updating KPIs, grid rows, and Spotlight automatically.

COMPARISON

When to Use "Read Log" vs "Start Real-time"

Criteria	Read Log (Historical Scan)	Start Real-time (2s Polling)
Primary Use Case	Auditing past shifts, yesterday's production, or offline rotated archives from disk.	Live shop-floor production monitoring; appends newly flushed lines every 2 seconds.
`From / To` Date Filter	Streams only `OTP_1.log` and rotated `OTP_1.N.log` intersecting the chosen window.	If `To` date is stuck in the past, the app prompts or clears the upper limit so live panels remain visible.
Changing Log Paths	Freely editable via text field or `OTP ...` / `MES ...` browse dialogs.	Click **Stop Real-time** before switching active machine log file paths.

NOTES

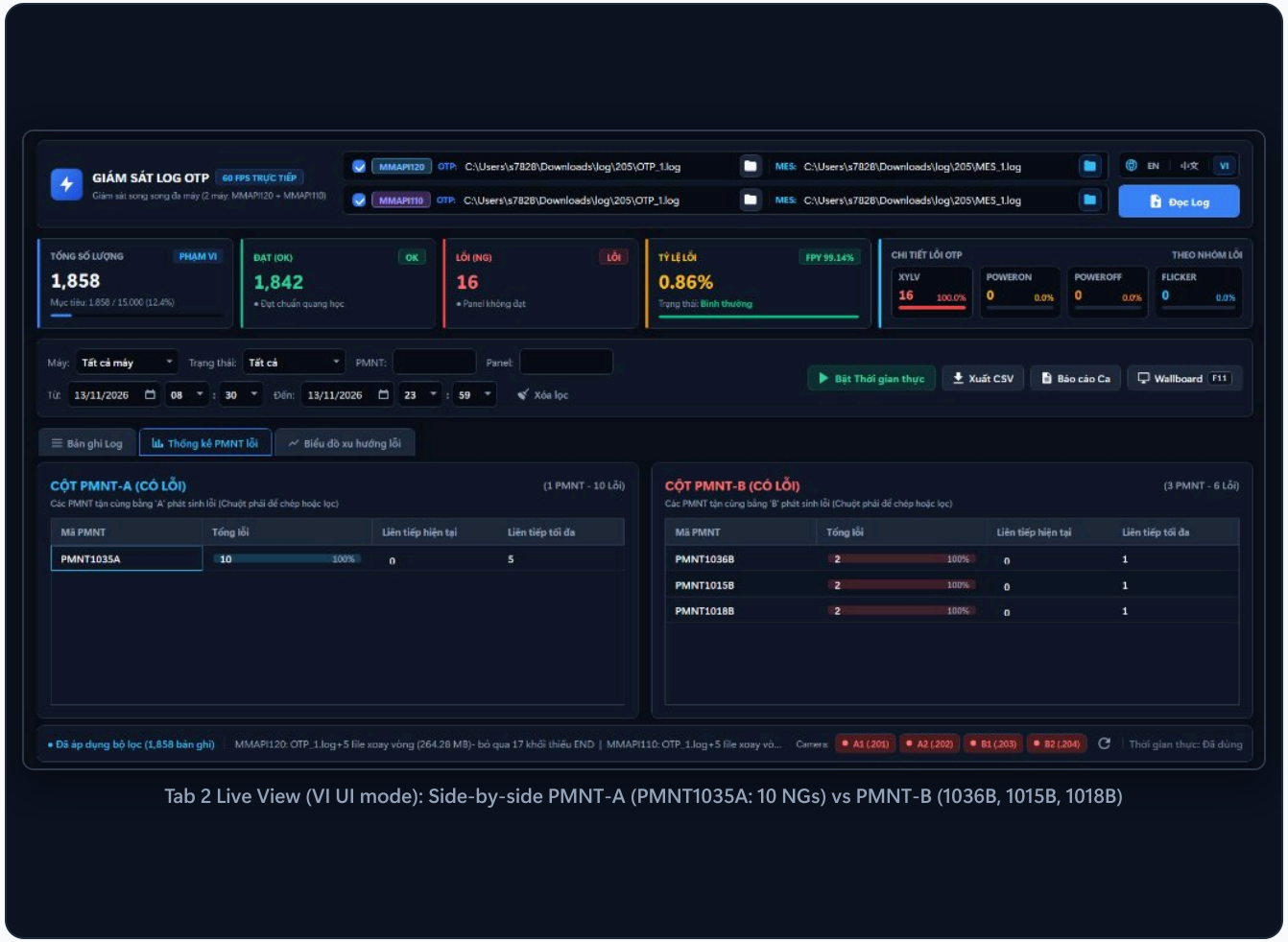
Interpreting Footer Diagnostics Messages

- `+5 rotated`: Automatically merged 5 historical rotated files (e.g. `OTP_1.1.log` .. `OTP_1.5.log`) into the active view.
- `skipped N blocks missing END`: Tester was mid-write during scan; incomplete trailing blocks are safely held and re-read on the next 2s cycle.
- `older trimmed`: RAM holds the latest 30,000 units for 60 FPS fluidity; narrow `From/To` dates and click **Read Log** to inspect older slices.
- **Camera Footer LEDs:** `Green` = SMB TCP 445 online; `Red` = Offline (click Probe button on the right to re-test).

TIP: Check both **Auto-scroll to latest record** and **Play sound on NG defect** above Tab 1 so line operators react immediately without constant screen watching.

7. Instant Filtering, Faulty PMNT Ranking & Shift Exports

Tab 1: Log Records • Tab 2: Faulty PMNT Statistics



Tab 2 Live View (VI UI mode): Side-by-side PMNT-A (PMNT1035A: 10 NGs) vs PMNT-B (1036B, 1015B, 1018B)

TAB 1 150ms Instant Filtering & Context Menu

- Filter immediately by *Machine*, *Status*, substring `PMNT` / `Panel`, and `From/To` time window.
- Right-click any row to *Copy Cell/Row*, *Filter PMNT/Panel*, or **Open Left/Right Cam Folder**.

TAB 2 Faulty PMNT Ranking (A / B / Other)

- **Automatic Side Separation:** Splits fixtures ending in `'A'`, `'B'`, and `Other` to isolate left/right dual-socket faults quickly.
- **3 Key Metrics:** Tracks *Defect Count*, *Current Streak*, and *Max Streak* to spot degraded pins or optical sockets.
- **Stop-Jig Alarm Threshold:** Hitting `3 consecutive NGs` fires a red Banner alarm; right-click any PMNT row to copy or isolate its logs.

EXPORT Manual CSV & Scheduled Shift Handover

- **Export CSV (Manual):** Exports filtered rows as UTF-8 CSV with OWASP CSV-Injection neutralization (`= + - @`) and standardized English headers for QA BI tools.
- **Shift Report (Manual & Auto):** Click **Shift Report** anytime, or auto-generates at `20:30` (Day) & `08:30` (Night) into `exports\`:
  - `ShiftReport_{yyyyMMdd}_{DAY|NIGHT}.csv`
  - `ShiftChart_{yyyyMMdd}_{DAY|NIGHT}.png`

> **TIP:** Double-click any cell in *Log Records* to copy its exact value, or press `Ctrl` + `C` to copy selected rows cleanly.

8. Interactive "Defect Trend Chart" Tab (3 Analytics Modes)

Tab 3 • 60 FPS Hardware DrawingVisual

MODE 1 **Pareto 80/20**

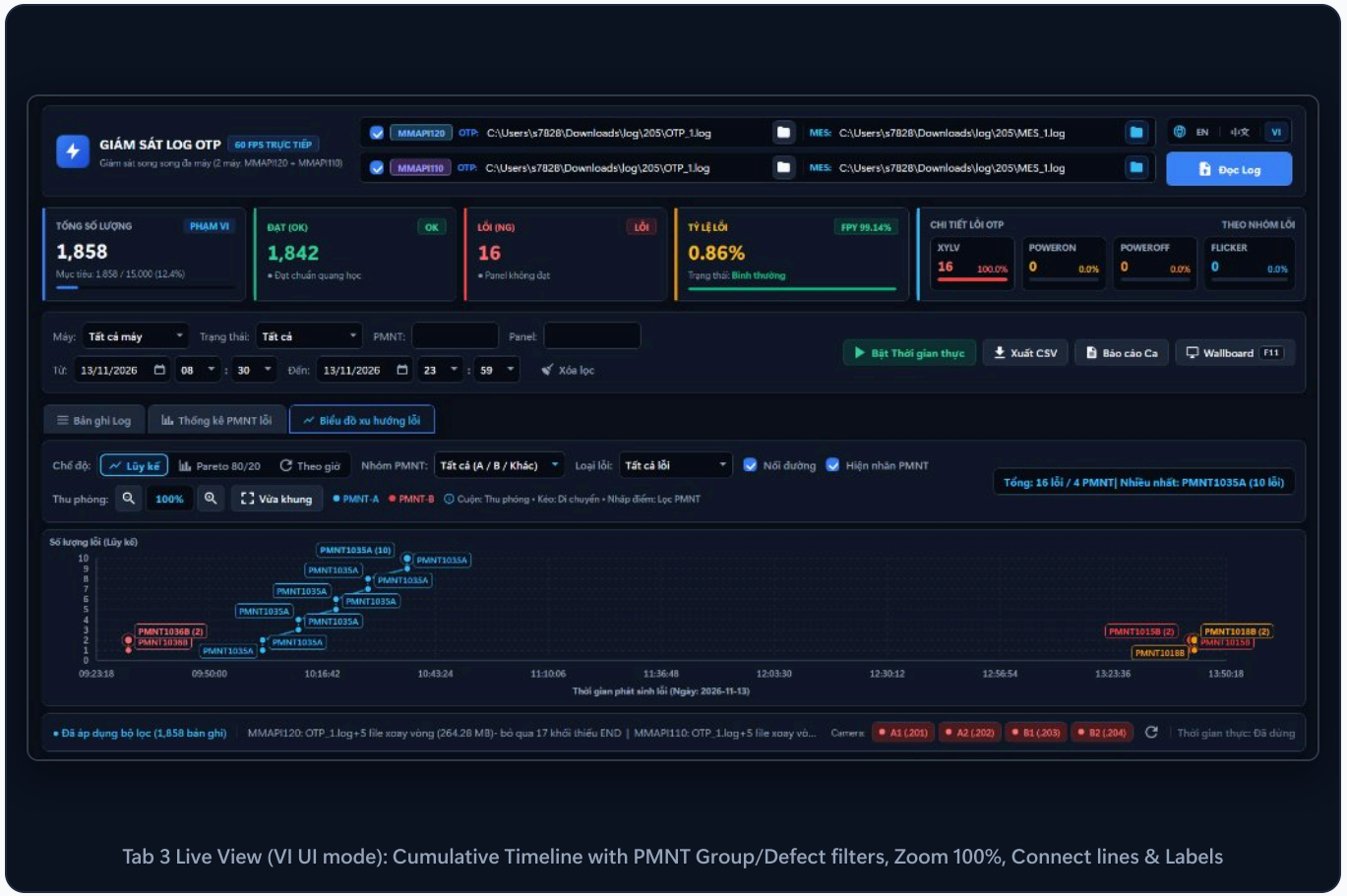
Ranks the **Top 10 defective PMNTs** with cumulative percentage curve and dashed **80% Cutoff** line to isolate the ~20% fixtures driving 80% of total NGs.

MODE 2 **Hourly Trend**

Dual-axis combo chart plotting **Throughput (Units/Hour)** bars against the **NG Rate %** trend line per shift hour, highlighting peak UPH.

MODE 3 **Cumulative Timeline**

Plots step-by-step cumulative defect growth over timestamps per PMNT code—ideal for spotting the exact moment a fixture starts drifting.



Tab 3 Live View (VI UI mode): Cumulative Timeline with PMNT Group/Defect filters, Zoom 100%, Connect lines & Labels

Interactive Mouse Controls & Chart Filters

- **Dedicated Filters:** Slice by *PMNT Group* (PMNT-A / PMNT-B / Other Only) and *Defect Type* (`XYLV` , `POWERON` , `POWEROFF` , `FLICKER`); toggle *Connect lines* & *Show PMNT labels*.
- **Zoom & Pan:** Scroll wheel or click `−` / `+` to zoom in on dense timestamps, drag with left mouse button to pan, or click **Fit Frame** for 100% reset.
- **Hover Data-Point Tooltip:** Displays full PMNT code, defect sequence #, exact timestamp, OTP error, machine, Panel ID, and BLU ID.
- **Left-Click Point:** Toggles isolation filter for that single PMNT directly inside the chart.
- **Right-Click Point:** Jumps straight to *Log Records* filtered by that PMNT to inspect MES specs & camera images.

9. Fullscreen Wallboard Mode (F11) & Live NG Spotlight

Designed for Overhead Line TVs



Wallboard F11 Live View (VI UI mode): Slide [1/4] Pareto 80/20 + Live NG Unit Spotlight with live MES specs

Activating & Controlling Wallboard Kiosk

- **Enter / Exit:** Press `F11` (or click **Wallboard**) to hide configuration chrome and expand edge-to-edge for TVs. Press `F11` or `ESC` to return.
- **4-View Auto-Rotate Carousel:** `[1/4]` Pareto 80/20 → `[2/4]` Hourly Trend → `[3/4]` Defect Timeline → `[4/4]` Top PMNT A/B/Other Ranking.
- **Carousel Pacing:** Toggle `15s / 30s` cycle speed, click **Auto-Rotate** to pause on any single slide, or hit **Next View** to advance manually.

KEY INNOVATION

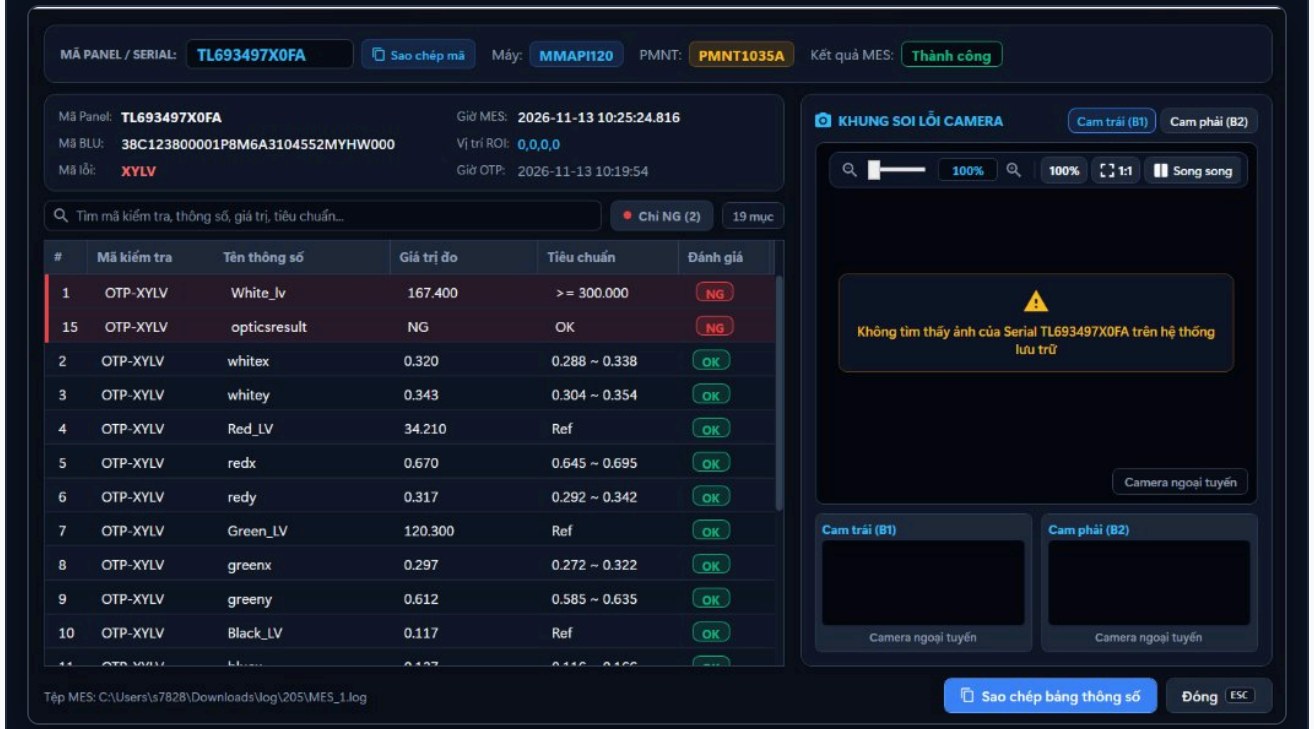
Live NG Unit Spotlight

Whenever any monitored machine flags an NG panel, the **LIVE NG UNIT SPOTLIGHT** card automatically pins the unit and renders:

- **Instant Unit Identity:** Panel serial, Machine, `PMNT` code, consecutive streak count, and primary OTP defect (`XYLV` , `POWERON` ...).
- **Async Dual-Cam Thumbnails:** Decodes Left & Right camera previews at `360 px` (prioritizing matching defect crops) without stalling the 2s tail loop.
- **Out-of-Spec MES Highlights:** Extracts failing optical metrics directly from `MES_1.log` side-by-side with the images.

10. MES Parameter Details & Camera Defect Viewer (MesDetailWindow)

Double-click Row or Click "View Details"



MesDetailWindow Live View (VI UI mode): NG Only filter active, 100%–500% zoom toolbar, 1:1 & Side-by-Side modes

LEFT PANE — MES Optical Parameter Spec Check

- Select **MES Run** if re-tested; search metrics, toggle **NG Only (N)**, or export TSV via **Copy Parameter Table**.

MES Parameter	Default Standard Spec	Eval
`White_lv` / `contrast`	`≥ 300.000` / `≥ 1050`	OK / NG
`whitex` / `whitey`	`0.288~0.338` / `0.304~0.354`	OK / NG
`cct` / `ntsc` / `flk`	`5500~8000` / `≥0.42` / `0~30`	OK / NG
`Red_LV`, `Green_LV`, `Blue_LV`	Optical reference measurements	Ref

RIGHT PANE — Camera Defect Viewer

- **Channel & Compare:** Switch **Left Cam** / **Right Cam** or enable **Side-by-Side** split comparison.
- **Defect Crop vs Overview:** Toggle **Crop** or click thumbnail strips inside `defectPicture/` below the viewport.
- **100%–500% Zoom & 1:1 Raw:** Wheel zoom / drag pan, double-click to fit `100%`; click **1:1** to stream raw `photo/Main*.png` on demand (capped at `4096 px`).

Press **ESC** to clear the MES search box or close the detail window.

11. Runtime Trilingual UI & External `config/settings.json`

Zero-Recompile Configuration

i18n **Instant Trilingual Switch (`EN | 中文 | VI`)**

Engineers across Vietnam, China, and international teams can toggle the entire interface language live from the header bar with **no restart required**:

EN	中文	VI
`Strings.en.xaml`	`Strings.zh.xaml`	`Strings.vi.xaml`

Standard factory tokens (`OTP` , `MES` , `PMNT` , `FPY` , `XYLV` , `OK` , `NG`) stay unified across all three dictionaries.

AUTO SAVE **File `config/user-state.json`**

- Operator UI preferences automatically persist to `config\user-state.json` :
- Active interface language (`en` / `vi` / `zh`).
- Per-machine enabled checkboxes (`MMAPI120` , `MMAPI110` ...) & custom log paths.
- Last selected `AutoScroll` , `SoundAlert` , and active analytics tab state.

EXTERNAL CONFIG **Edit `config\settings.json` via Notepad**

```
{
  "machines": [
    {
      "name": "MMAPI120", "enabled": true, "color": "#74B9FF",
      "logPath": "\\\\192.168.3.206\\c\\GoArms\\logs\\OTP_1.log",
      "mesLogPath": "\\\\192.168.3.206\\c\\GoArms\\logs\\MES_1.log",
      "images": [
        { "root": "\\\\192.168.3.203\\e\\LOG", "code": "B1" },
        { "root": "\\\\192.168.3.204\\e\\LOG", "code": "B2" }
      ]
    }
  ],
  "wallboard": {
    "warningNgRatePercent": 2.0, "criticalNgRatePercent": 5.0,
    "shiftTargetUnits": 15000, "carouselIntervalSec": 15
  },
  "autoShiftExport": {
    "enabled": true, "dayShiftStart": "08:30", "nightShiftStart": "20:30",
    "dayShiftTime": "20:30", "nightShiftTime": "08:30"
  },
  "mesSpecs": {
    "White_lv": { "min": 300.0, "spec": "≥ 300.000" },
    "contrast": { "min": 1050.0, "spec": "≥ 1050" }
  }
}
```

JSON RULE: Every backslash in UNC SMB paths (`\\192.168 ...`) must be escaped twice (`\\\\192.168 ...`) inside JSON. On syntax errors, the app safely falls back to defaults and displays a non-blocking amber warning bar.

12. Single-File Factory Deployment & Shortcut Cheat Sheet

Zero Pre-Installed Runtime Needed

ROLLOUT Clean Windows 10/11 x64 Station Deployment

- **Automated Packaging:** Run `.\build-package.ps1` at the repo root to verify all xUnit tests, build compressed Single-File binaries, and package `OTPSoftware_v1.0.0_YYYYMMDD_win-x64.zip`.
- **Copy, Unzip & Run:** Extract the ZIP onto any shop-floor PC (e.g. `D:\Apps\OTP_Software\`). The entire .NET 10 Desktop Runtime is bundled inside `OTPSoftware.App.exe`.
- **Working-Dir Independent:** Whether launched by double-click, Desktop shortcut, or CLI, `AppContext.BaseDirectory` probing reliably resolves external `config\settings.json` beside the `.exe`.

```
OTP_Software\
├── OTPSoftware.App.exe       # Standalone executable (embedded .NET 10 Runtime)
├── README.txt                # Quick shop-floor manual & changelog
├── config\settings.json      # Editable machine/camera IPs outside bundle
├── logs\gui-error.log        # Self-rotating crash diagnostics (created on demand)
└── exports\                  # ShiftReport_*.csv & ShiftChart_*.png outputs
```

SMOKE CHECK: Run `OTPSoftware.App.exe --validate-startup` in PowerShell/CMD — returning `STARTUP_VALIDATION_SUCCESS` confirms the station is production ready.

CHEAT SHEET Hotkeys & Mouse Interactions

Key / Action	Active Screen	Behavior Executed
`F11`	Main Dashboard	Toggle fullscreen TV **Wallboard Kiosk** mode on/off.
`ESC`	Wallboard / Detail	Exit Wallboard; clear MES parameter search or close `MesDetailWindow`.
`Enter`	Top Path Inputs	Commit edited log file path and trigger disk log reload.
`Ctrl` + `C`	Log Records Grid	Copy all selected records cleanly to Clipboard (Excel-ready TSV).
Double-Click Cell	Log Records Grid	Instantly copy exact cell value (Panel, BLU, PMNT, Status...).
Double-Click Row	Log Records Grid	Launch **MES Parameter Details & Camera Defect Viewer**.
Right-Click	Grid / PMNT / Chart	Context actions: Open Left/Right Cam Folder, Filter PMNT, Jump to Logs.
Wheel / Drag	Chart / Cam Viewer	Zoom & pan dense charts or defect crops; double-click photo to reset 100%.

13. Shop-Floor Troubleshooting & Version 1.0 Sign-Off

Diagnostic Log: logs/gui-error.log

TROUBLESHOOTING

Quick Shop-Floor Field Diagnostics Guide

Symptom / Status Alert	Typical Root Cause	Verification & Corrective Action
Red Footer LED / "Camera offline"	Camera IPC powered off, LAN drop, or SMB TCP port 445 read share inaccessible.	Test ping (e.g. `192.168.3.203`) and browse `\\192.168.3.203\e\LOG` in Explorer. Circuit Breaker recovers automatically within 10s (or click Probe).
"Panel not found in MES"	`mesLogPath` points to the wrong host, or payload sits in an older rotated file.	Verify the matching `MES ...` button per machine (`MMAPI110` vs `MMAPI120`) or update `mesLogPath` inside `config\settings.json`.
"Folder not found" for Panel	Unit image directory archived/deleted, or folder name omits expected Serial token.	Verify folder matching `*[Serial]*_[Station]` exists under date folders or root path on the camera share.
Amber Warning Banner on `settings.json`	JSON syntax error: missing comma, brace, or unescaped backslash `\\` from Notepad edit.	Inspect braces and double backslashes `\\\\` in `config\settings.json`, save UTF-8, and relaunch the app.
Unexpected Application Exception	Transient workstation I/O interruption or unhandled desktop UI event.	Attach `logs\gui-error.log` (safe size-capped rotating log with full stack trace and line numbers) when escalating to Engineering.

RELEASE SIGN-OFF • OTP_SOFTWARE V1.0.0

Production Ready for 24/7 OTP / AOI Shop-Floor Lines

Clean Architecture (.NET 10 / C# 14), verified by the automated **xUnit** regression suite, headless startup smoke check (`STARTUP_VALIDATION_SUCCESS`), and distributed as a self-contained **win-x64** Single-File package.

Repository: `Danielz2301/OTP_Software`
Commit: `Version1.0` (Clean Single Root)
Package Script: `.\build-package.ps1`